

$$2. \quad m = 100g \quad r_1 = 70cm \quad F = 1,5N$$

$$r_2 = 30cm$$

$$F = m \frac{v_1^2}{r_1} \rightarrow v_1 = \sqrt{\frac{F \cdot r_1}{m}} = \underline{3,24 \text{ m/s}}$$

$$L_1 = r_1 m v_1 = L_2 = r_2 m v_2$$

$$v_2 = \frac{r_1 v_1}{r_2} = \underline{7,56 \text{ m/s}}$$

$$\Delta W = \frac{1}{2} m (v_2^2 - v_1^2) = \underline{2,33 \text{ J}}$$

$$3. \quad A: m_A \cdot a = F_{SA} - m_A \cdot g \quad B: F_{SC} - F_{SA} - m_B \cdot g \cdot \mu = m_B \cdot a$$

$$C: m_C \cdot a = m_C \cdot g - F_{SC}$$

$$A+B: (m_A + m_B) \cdot a = F_{SC} - m_A \cdot g - m_B \cdot g \cdot \mu$$

$$F_{SC} = (m_A + m_B) \cdot a + m_A \cdot g + m_B \cdot g \cdot \mu$$

$$(A+B)+C: m_C \cdot g - m_C \cdot a = (m_A + m_B) \cdot a + m_A \cdot g + m_B \cdot g \cdot \mu$$

$$a) \quad a = \frac{m_C \cdot g - m_A \cdot g - m_B \cdot g \cdot \mu}{m_A + m_B + m_C} = \underline{0,3924 \text{ m/s}^2}$$

$$b) \quad F_{SA} = m_A \cdot a + m_A \cdot g = \underline{61,26 \text{ N}}$$

$$F_{SC} = m_C \cdot g - m_C \cdot a = \underline{84,8 \text{ N}}$$

$$c) \quad \mu = \frac{m_C \cdot g - m_A \cdot g}{m_B \cdot g} = \underline{0,3}$$

4. Geg: $m = 8,9 \cdot 10^{22} \text{ kg}$ $r = 422 \cdot 10^3 \text{ km}$, $T = 1,77 \text{ d}$
 $\geq 152,93 \cdot 10^3 \text{ s}$
 Ges: M von Jupiter

$$F = \gamma \cdot \frac{m \cdot M}{r^2}$$

$$F = m \frac{v^2}{r}$$

$$v = \frac{2\pi r}{T} = \frac{2\pi \cdot 422 \cdot 10^6 \text{ m}}{152,93 \cdot 10^3 \text{ s}} = ~~17~~ 338 \text{ m/s}$$

$$F = \frac{m \cdot v^2}{r} = \frac{8,9 \cdot 10^{22} \cdot v^2}{422 \cdot 10^6 \text{ m}} = 63,4 \cdot 10^{21} \text{ N}$$

$$M = \frac{F \cdot r^2}{\gamma \cdot m} = \underline{\underline{1,9 \cdot 10^{27} \text{ kg}}}$$

5. Geg: $m = 5 \text{ kg}$ $r = 0,2 \text{ m}$ schnur: 2 m
 $F_s = 100 \text{ N}$ Achse: $0,02 \text{ m}$

a) $U_A = 2\pi r = 12,57 \text{ cm}$

$$n = \frac{2 \text{ m}}{0,1257 \text{ m}} = \underline{\underline{15,915}}$$

b) $M = I \cdot \alpha \Rightarrow F \cdot r = \frac{1}{2} m (r_1^2 + r_2^2) \cdot \alpha$

$$\alpha = \frac{r_A \cdot F}{\frac{1}{2} m (r_1^2 + r_2^2)} = \frac{0,02 \cdot 100 \text{ N}}{0,5 \cdot 5 (0,17^2 + 0,2^2)} = \underline{\underline{11,61 \text{ s}^{-2}}}$$

c) ① $E_{\text{kin}}^{\text{rot}} = \frac{1}{2} I_s \cdot \omega^2 = \frac{1}{2} I_s \cdot 48,187^2 = \underline{\underline{200 \text{ J}}}$

③ $\omega = \alpha \cdot t \rightarrow 11,61 \text{ s}^{-2} \cdot t = 48,187 \text{ s}^{-1}$

② $\varphi = \frac{1}{2} \alpha t^2 \rightarrow t = \sqrt{\frac{2 \cdot \varphi}{\alpha}} = \sqrt{\frac{2 \cdot 100}{11,61 \text{ s}^{-2}}} = \underline{\underline{4,15 \text{ s}}}$

④ $\varphi = 15,915 \cdot 2\pi = \underline{\underline{100}}$